Let $x(\tau)$ denote the susceptible fraction, $y_1(\tau)$ the infected fraction of strain 1, and $y_2(\tau)$ the infected fraction of strain 2. The dimensionless two-strain system is

$$\frac{dx}{d\tau} = \varepsilon(1-x) - \mathcal{R}_{0,1}xy_1 - \mathcal{R}_{0,2}xy_2,$$

$$\frac{dy_1}{d\tau} = (\mathcal{R}_{0,1}x - 1)y_1,$$

$$\frac{dy_2}{d\tau} = (\mathcal{R}_{0,2}x - 1)y_2,$$

where ε is the demographic time-scale parameter, and $\mathcal{R}_{0,1}$, $\mathcal{R}_{0,2}$ are the basic reproduction numbers of strain 1 and strain 2.

The boundary-layer thresholds are

$$y_1^\star = \varepsilon\left(1 - \frac{1}{\mathcal{R}_{0,1}}\right),$$

$$y_2^\star = \varepsilon\left(1 - \frac{1}{\mathcal{R}_{0,2}}\right),$$

assuming

$$\mathcal{R}_{0,1} > 1, \qquad \mathcal{R}_{0,2} > 1.$$

We consider the case where strain 1 has established a major epidemic and then declines into its boundary layer.

Let τ_1 be the dimensionless time at which strain 1 enters its boundary layer. Define

$$x_a = x(\tau_1),$$

$$y_1(\tau_1) = y_1^\star,$$

$$y_{2,a} = y_2(\tau_1),$$

and assume that strain 2 has not yet entered its boundary layer:

$$y_{2,a} > y_2^\star.$$

Define the local dimensionless time

$$s = \tau - \tau_1.$$

Thus $s = 0$ corresponds to the moment when strain 1 enters the boundary layer, and

$$x(0) = x_a,$$

$$y_2(0) = y_{2,a}.$$

During the stage where strain 1 is rare and strain 2 is resident, we use the approximation

$$\varepsilon(1-x) \ll \mathcal{R}_{0,2}xy_2,$$

and neglect the feedback of strain 1 on the susceptible pool. Therefore the resident strain-2 trajectory satisfies

$$\frac{dx}{ds} = -\mathcal{R}_{0,2}xy_2,$$

$$\frac{dy_2}{ds} = (\mathcal{R}_{0,2}x - 1)y_2.$$

Dividing the second equation by the first gives

$$\frac{dy_2}{dx} = \frac{dy_2/ds}{dx/ds} = \frac{(\mathcal{R}_{0,2}x - 1)y_2}{-\mathcal{R}_{0,2}xy_2}.$$

Thus

$$\frac{dy_2}{dx} = -1 + \frac{1}{\mathcal{R}_{0,2}x}.$$

Integrating from $(x_a, y_{2,a})$, we obtain

$$y_2(x) - y_{2,a} = \int_{x_a}^{x} \left(-1 + \frac{1}{\mathcal{R}_{0,2}z}\right) dz,$$

where z is the integration variable. Hence

$$y_2(x) = y_{2,a} + x_a - x + \frac{1}{\mathcal{R}_{0,2}} \ln \frac{x}{x_a}.$$

Let x_b denote the susceptible fraction when strain 2 enters its boundary layer. It is determined by

$$y_2(x_b) = y_2^\star,$$

that is,

$$y_{2,a} + x_a - x_b + \frac{1}{\mathcal{R}_{0,2}} \ln \frac{x_b}{x_a} = y_2^\star.$$

Define

$$C_2 = y_{2,a} + x_a - y_2^\star.$$

Then x_b satisfies

$$x_b - \frac{1}{\mathcal{R}_{0,2}} \ln \frac{x_b}{x_a} = C_2.$$

Using the Lambert W function, the relevant late-epidemic root is

$$x_b = -\frac{1}{\mathcal{R}_{0,2}} W_0\left(-\mathcal{R}_{0,2}x_a \exp(-\mathcal{R}_{0,2}C_2)\right),$$

where W_0 is the principal branch of the Lambert W function.

Now compute the probability that strain 1 goes extinct before strain 2 enters its boundary layer.

By Kendall's finite-time formula for a time-inhomogeneous birth-death process, if the per capita birth rate is $b(s)$, the per capita death rate is $d(s)$, and

$$H(s) = \int_0^s [b(u) - d(u)]\, du,$$

then the probability that a process starting from one individual is still nonzero at time s is

$$S(s) = \frac{\exp(H(s))}{1 + \int_0^s b(v) \exp(H(v))\, dv}.$$

For strain 1 in the strain-2 resident environment,

$$b_1(s) = \mathcal{R}_{0,1}x(s),$$

$$d_1(s) = 1.$$

Therefore

$$H_1(s) = \int_0^s [\mathcal{R}_{0,1}x(u) - 1]du,$$

and the probability that one strain-1 infected individual is still present at time s is

$$S_1(s) = \frac{\exp(H_1(s))}{1 + \int_0^s \mathcal{R}_{0,1}x(v)\exp(H_1(v))\,dv}.$$

We now rewrite this in terms of x. From

$$\frac{dx}{ds} = -\mathcal{R}_{0,2}xy_2,$$

we obtain

$$ds = -\frac{dx}{\mathcal{R}_{0,2}xy_2(x)}.$$

Thus, when the susceptible fraction decreases from x_a to x, define

$$\widehat{H}_1(x) = \int_x^{x_a} \frac{\mathcal{R}_{0,1}z - 1}{\mathcal{R}_{0,2}z\,y_2(z)}\,dz,$$

where

$$y_2(z) = y_{2,a} + x_a - z + \frac{1}{\mathcal{R}_{0,2}}\ln\frac{z}{x_a}.$$

The denominator integral in Kendall's formula becomes

$$B_1(x) = \int_x^{x_a} \frac{\mathcal{R}_{0,1}}{\mathcal{R}_{0,2}y_2(z)}\exp(\widehat{H}_1(z))\,dz.$$

Therefore, when the strain-2 resident trajectory reaches susceptible fraction x, the survival probability of one strain-1 infected individual is

$$S_1(x) = \frac{\exp(\widehat{H}_1(x))}{1 + B_1(x)}.$$

At the moment strain 2 enters its boundary layer, $x = x_b$. Hence the probability that one strain-1 infected individual has gone extinct before that moment is

$$1 - S_1(x_b).$$

The number of strain-1 infected individuals at its boundary-layer entrance is

$$m_1 = Ny_1^\star,$$

where N is the total population size. Under the branching-process approximation, the descendant trees of the m_1 initial infected individuals are independent. Therefore the probability that strain 1 goes extinct before strain 2 enters its boundary layer is

$$P_{m_1}^{1|2} = [1 - S_1(x_b)]^{m_1},$$

where the superscript $1 \mid 2$ means that strain 1 is rare and strain 2 is resident.